



1. Define Yellow hat? 2 Marks



The Yellow Hat

The Yellow Hat symbolizes brightness and optimism.

2. Define Green hat 2 marks



The Green Hat

The Green Hat focuses on creativity: the possibilities, alternatives and new ideas.

3. Define blue hat 2 marks



The Blue Hat

The Blue Hat is used to manage the thinking process.

4. Define Red hat 2 marks



The Red Hat

The Red Hat signifies feelings, hunches and intuition.

5. umbrella term

The names may be different but umbrella term for it is "Early Childhood Education."

6. negative reinforcement 2 marks

Negative reinforcement occurs when something already present is removed (taken away) as a result of a person's behavior, creating a favorable outcome for that person. Basically, when a person's behavior leads to the removal of something that was unpleasant to that person then negative reinforcement is occurring.

Reinforcement theory of motivation says that children's behavior can be modified through positive or negative reinforce.

- **Negative reinforcement:** the taking away of an aversive stimulus to increase certain behavior or response.

Example: Putting ointment on a bug bite to soothe an itch. If using ointment on bug bites increases, the removal of an itch is a negative reinforcer.

7. what is measuring? 2 marks

Identifying and ordering lengths and then areas, volume, weights, temperature and speeds.

8. Physiological needs 2 marks

Physiological needs:

The definition of physiological needs is quite obvious; they are the physical requirements for human survival. If these requirements are not met, the human body cannot function properly, and will ultimately fail. Physiological needs are thought to be the most important; they should be met first. Therefore, a human lacking food, love, esteem, or safety would consider the greatest of his/her needs to be food.

9. what is project learning? 2 marks

Project Learning:

- According to research, if projects are structured properly, they allow the learner or groups of learners to be immersed in one big idea, or to work towards one common goal.
- The project challenges learners to use a variety of skills and intelligence and allows them to be involved in meaningful activity.
- Projects tend to be multidimensional and they encourage learners to extend their understanding of the content beyond the classroom and into real world.
- It is important for the teacher to consider students' developmental level when designing projects.

10. cooperative learning 3 marks

Cooperative learning is the process of breaking a classroom of students into small groups so they can discover a new concept together and help each other learn.

11. Adolescent brain Changes 3 marks

- Research has now determined that remarkable changes occur in the brain during the second decade of life.
- The understanding that adolescence is a time of profound brain growth and change is contrary to long-held ideas that the brain was mostly fully "formed" by the end of childhood.
- Like a computer, the maturing brain grows "circuits"-neural connections-that can perform several tasks simultaneously and with ever-greater efficiency.
- Dopamine inputs to the PFC-a chemical messenger critical for focusing attention when necessary to choose between conflicting options- grow dramatically during adolescence.

12. Conclusion in scientific method 3 marks

Conclusion

Hassan rejects his hypothesis, but decides to re-test using sugar amounts between 50g. and 100g.

13. De Bono Six Thinking Hats 5 marks

De Bono Six Thinking Hats: The term *Six Thinking Hats* is used to describe the tool for group discussion and individual thinking. "Six Thinking Hats" and the associated idea parallel thinking provide a means for groups to plan thinking processes in a detailed and cohesive way, and in doing so to think together more effectively.

The Six Thinking Hats (or modes)



The White Hat

The White Hat calls for information known or needed.



The Red Hat

The Red Hat signifies feelings, hunches and intuition.



The Black Hat

The Black Hat is judgment -- the devil's advocate or why something may not work.



The Yellow Hat

The Yellow Hat symbolizes brightness and optimism.



The Green Hat

The Green Hat focuses on creativity: the possibilities, alternatives and new ideas.



The Blue Hat

The Blue Hat is used to manage the thinking process.

14. Is the relationship between experiment and discovery learning. How? 5Marks

- **Experiments:** Definitely in experiments the students are discovering, not only at the end but in the process of experiment as well. E.g. sometimes students may discover that the material used in the experiment is not of a good quality. Students discover knowledge for themselves.

15. Types of goal structure?

- **Goal structure:** It is a way that goals specify the degree of interdependence sought among students.

There are three different types of goal structures:

- individualistic,
- competitive
- cooperative.

16. hypothesis

Hypothesis

The hypothesis is an educated guess about the relationship between the independent and dependent variables.

Reading is involved behind the educated guess. Experts opinions are also involved in it.

17. goal structure

Goal stucture: It is a way that goals specify the degree of interdependence sought among students.

18. Define Reinforcement theory its types explain 5

Reinforcement theory emphasizes the importance of individuals responding to environmental events and extrinsic reinforcements.

Environmental factors are important to reinforce certain behavior. E.g. if you want to reinforce a behavior among your students that they must do their homework. You can reinforce this behavior among students by giving them rewards.

To inforce behavior positively among students is a positive reinforcement.

Negative reinforcement occurs when something already present is removed (taken away) as a result of a person's behavior, creating a favorable outcome for that person. Basically, when a person's behavior leads to the removal of something that was unpleasant to that person then negative reinforcement is occurring.

Reinforcement theory of motivation says that children's behavior can be modified through positive or negative reinforce.

- **Positive reinforcement:** the adding of an appetitive stimulus to increase a certain behavior or response.
Example: Father gives candy to his daughter when she picks up her toys. If the frequency of picking up the toys increases or stays the same, the candy is a positive reinforcer.
- **Positive punishment:** the adding of an aversive stimulus to decrease a certain behavior or response. Example: Mother yells at a child when running into the street. If the child stops running into the street the yelling is positive punishment.
- **Negative reinforcement:** the taking away of an aversive stimulus to increase certain behavior or response.
Example: Putting ointment on a bug bite to soothe an itch. If using ointment on bug bites increases, the removal of an itch is a negative reinforcer.
- **Negative punishment (omission training):** the taking away of an appetitive stimulus to decrease a certain behavior.
Example: A teenager comes home an hour after curfew and the parents take away the teen's cell phone for two days. If the frequency of coming home after curfew decreases, the removal of the phone is negative punishment.

19. syntax of discovery learning 5 marks

Syntax of discovery learning: (Steps)

There are 9 steps of discovery learning, and discovery learning should be done according to these nine steps:

1. **Select an activity.** Choose an activity that does not have just one correct answer. Roleplaying, creating sculptures, observing characteristics of subjects, or searching for or classifying similar items all work well. (Select an activity that ensures discovery learning in it.)
2. **Gather material.** Plan enough materials for each learner to repeat the activity at least once.
3. **Stay focused.** Avoid learning tangents that may be interesting but will keep the learner away from finishing the project. (Whenever discovery learning is planned, there are many

interesting tasks for the students. Sometimes students attention is diverted, it should not happen. Teacher needs to ensure that students are focused.)

4. **Use caution.** While the idea of discovery learning is for the instructor to step back and observe allowing the child to work independently, be sure that safety is observed.
5. **Plan extra time.** Understand that children working on their own will most likely take longer than they would with an adult moving them from step to step. Also be sure to plan time for repeated activities in case there is a failure or other reason to repeat the activity.
6. **Record process and results.** Include in the activity a requirement for older children to record their procedure and results. For young children guide, assist, or model record keeping.
7. **Feedback and review.** After an activity is completed and before it is repeated a second time (if needed), discuss the activity and its outcome with the child. Use the records which were kept to assist during this step.
8. **Try again.** Have the child repeat the activity if necessary. Give assistance and guidance as necessary.
9. **Plan for more discovery learning activities.** Think over how this activity worked for the child. As you plan more discovery activities take the answers to these questions into consideration. What went well? What could have gone better? How can any problem areas be corrected or alleviated?

20. Brain development activities 5 marks

Brain Development activities:

Morning greeting: There are brain cells in the minds of the children, but the connection between these cells are missing. This morning greeting provides the environment which helps in building the connections. The important role in this regard is of language.

Finger play in pre-school: By a couple of month of age, babies can process the emotional colors of language (prosody).

- In fact, toddlers can memorize nursery rhymes because rhymes have prosody. Finger play is important in developing connections in brain. As the preschool teacher raises her voice an octaves draws out her vowels, the child's brain responds by sending even more chemical and electrical impulses across the synapses.

Story time needs to be an integral part of early years of education. Because through story, a child gets language development, social development and social interaction.

Free play: It is not necessary that every school follows teacher-oriented activities i.e. the teacher always tells a story. But a teacher must give the students free time so that they can do something in groups either they could play together. **Research says, "During free play, pre-school children interact with one another. As they communicate, whether through beginning language or more sophisticated use of words, the neurons in their brains are making more connections, critical for reinforcing learning."**

Snack & circle time activity: This activity is important because the children can interact with one another more easily while having their snacks. Their social interaction is more at this time and most importantly, they learn sharing here; sharing of ideas in adult life comes only if someone experiences these things in early childhood. Sharing at early ages starts with little things e.g. snacks, toffees, chips, biscuits, lunch etc.

If the children do not share these little things at early ages then they would not share their ideas in their future lives. So, snack time activity is very helpful in learning of vocabulary, sentence construction, and their social development.

- **Circle time activity:** As the early childhood teacher focuses her attention on each individual child in the large group activity, the child must think about the topic for the day. The child's brain will be active as she/he retrieves from memory something special.

Art activity: Art is very important in early childhood education as it helps in brain development

21. Cooperative learning in detail 5 marks

Why cooperative learning is essential?

There are many reasons behind providing the students with cooperative learning. A lot of research is being done on it. We human beings are social animals and we are dependent on each other. A research was conducted in which 45 already conducted researches were analyzed about cooperative learning. 37 out of 45 researches say that if students work in groups than their academic achievement is higher. Other remaining 8 researches concluded that if students work in groups or individually, their achievement remains the same. This was an experimental research in which one was control group and the other was experimental group. Control group was given individual tasks while experimental group was provided with cooperative learning tasks. And at the end, 37 researches shows that the experimental group, who worked using cooperative learning styles performed much better than the controlled group.

The other remaining 8 researches concluded that if students work in groups or individually, their achievement remains the same.

There is no research which says that cooperative learning has a negative effect. If more than 80% of the researches say that cooperative learning helps in academic achievement of a student than essentially cooperative learning tasks must be provided to them.

Cooperative learning:

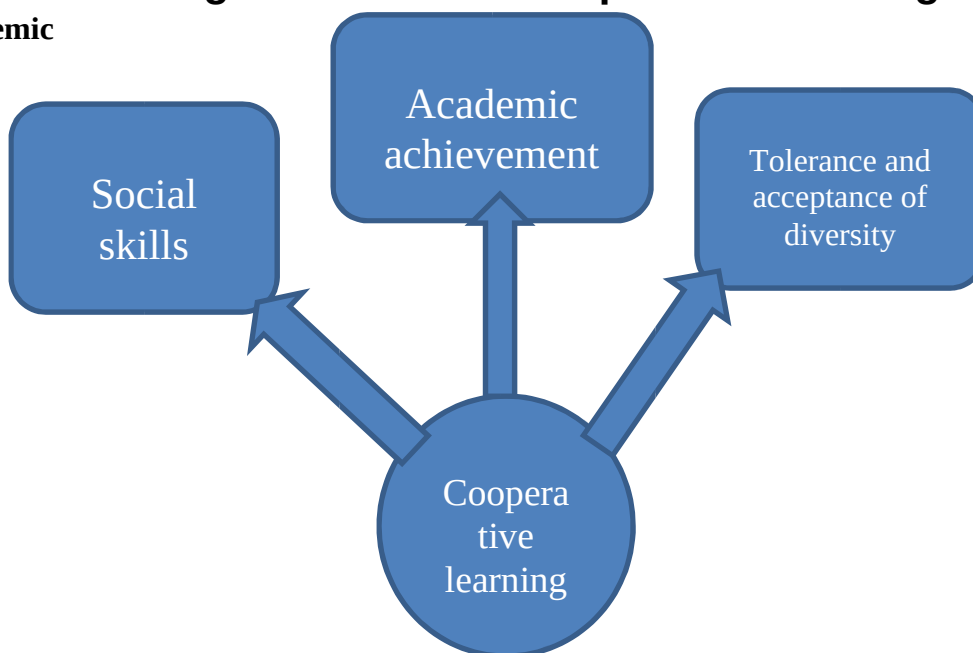
- It is a model which requires student cooperation and interdependence in its task, goal and reward structure.
- The cooperative learning model is characterized by cooperative task, goal and reward structures. Further students in cooperative learning situations are encouraged and/or required to work together on a common task, and they must coordinate their efforts to complete the task.
- Cooperative learning lessons can be characterized by the following features:

1. Students work in teams to master learning goal.
2. Teams are made up of high, average and low achieving students.
3. Whenever possible, teams include a racial, cultural and gender mix.
4. Reward systems are oriented to the group as well as the individual.

Many classrooms are observed where seating arrangement is in such a way where students can cooperate with each other. In fact, the students are not doing cooperative tasks. They are performing individually. They are sitting in groups but they are not cooperating with each other. Tasks must be cooperative in which each student should actively participate. Make sure that cooperative learning is challenging enough and could not be done individually.

Goals/Learning outcomes for cooperative learning:

Academic



The cooperative learning model was developed to achieve at least three important instructional goals: academic achievement, tolerance and acceptance of diversity, and social skill development.

Social skills: Communication is the most important skill

22. what is jigsaw technique?

in jigsaw technique teacher is one who select the text. For better management teacher may present the information.

Jigsaw technique has two parts in first part children read the same text they are called home groups. In second part children of different home groups share their information. Problem in jigsaw technique is that when children do not understand the text in home groups they will not be able to present it in jigsaw groups. Understanding in home group is very much important.

Assigning different roles is a good thing in jigsaw technique a coach and a checker can be helpful in understanding the text. We should go for the second option. In management the most important thing is assigning the tasks.

23. state three names used by different countries for pre-schooling?

The names that different countries use for pre-schooling or early childhood education is: -

In British English, Nursery School or simply “nursery.” -

In the United States pre-school and pre K are used.

- In Pakistan, there are different names to be used in private sector. i.e. Montessori education, kindergarten education, play group, toddler or kids’ education etc.
- In public sector, the term mostly used is “Kachi Class”

24. functions of adaptation?

Its functions are refine and problem. One a skill is achieved students should be able to relate or refine that skill to their practical life. Performance includes find connections and innovations.

4 MAT model is very much comprehensive. Problem solving skill is the function of right brain.

Planning of lesson must be based on the above four points.

Assessing students learning

To assess the students according to the define objectives is the simple method. But if we use 4 MAT model, we have to assess the students on all four points(meaning, concepts, skills and adaptation). We have to assess the students according to our teaching models that we are using.

25. Unguided inductive inquiry 5

Unguided inductive inquiry:

Students investigate topic-related questions that are student formulated through student designed/selected procedures. When an activity is evaluated for its level of inquiry, a simple table establishing what is given to the learner determines at which level of inquiry the given activity resides.

Unguided inductive inquiry:

Here you will study what unguided inductive inquiry is? And how is it different from guided inquiry.

- During guided inductive inquiry, the teacher plays the key role in asking questions, prompting responses and structuring the material and situations.
- Once the class has mastered the techniques of guided inductive inquiry, teacher can introduce or allow for student-initiated situations for examining data, objects and events. Because the teacher’s role is minimized, the students’ activity increases.

26. Guided inductive inquiry:

Students investigate a teacher-presented question using student designed/selected procedures.

What would you do as a future teacher to enhance intrinsic motivation in your students?

Motivating students to become intrinsically motivated is no easy task for teachers. To some students, they enjoy learning. But for the others, this is where the challenge really comes in. To facilitate material to students "dry" would totally be out of my agenda. I would try to teach to my students to the point where the material becomes alive within them. I to make their minds grow in curiosity of the subject so continual learning can take place. I would do this by means of relating material to what interests the students. I would also put as much energy into my teaching

whether it be in my tone of voice, body language, appearance, or all of the above. One thing that I'd like to do also is at the end of my class period, I would try to always close my lessons leaving my students with a question to ponder on dealing with the subject material that was taught. That way in their spare time, their minds will continue to ponder on the lesson and question at hand.

Guided Inquiry is carefully planned, closely supervised targeted intervention of an instructional team of school librarians and teachers to guide students through curriculum based inquiry units that build deep knowledge and deep understanding of a curriculum topic, and gradually lead towards independent learning.

Guided Inquiry is grounded in a constructivist approach to learning, based on the Information Search Process developed by Kuhlthau, for developing students’ competence with learning from a variety of sources while enhancing their understanding of the content areas of the curriculum.

27. Steps for GII (Guided Inductive Inquiry) 5 marks

Steps	Examples:
1. Decide on the generalization(s) students should make during a particular unit of study	- Humans need balanced diet to keep good health. - Shortage of minerals can cause serious illness. - Galileo was condemned by the inquisition because he questioned the accepted facts.
2. Organize the learning activities and materials to expose strands and generalizations to students.	- Cases representing different diseases caused by imbalanced diet. - Cases representing different diseases caused by shortage of minerals. - Analysis of Galileo’s work and the religious rule of that time.
3. Ask students to write summary of the content that will form basis of the generalizations.	Students summarize the read cases or texts.
4. Ask students to identify sequences or patterns of events, objects, or other data in the content.	
5. Ask students to summarize these sequences or patterns in one sentence.	
6. Ask students to offer proof that their statement is a generalization	

by applying it to other events, objects or data.

28. Moslow’s hierarchy 5

This level of need refers to what a person's full potential is and the realization of that potential. Maslow describes this level as the desire to accomplish everything that one can, to become the most that one can be. However, this is a broad definition of the need for self-actualization; when applied to individuals, the need is specific. For example, one individual may have the strong desire to become an ideal parent. In another, the desire may be expressed athletically.

29. Inductive methods 3

Inductive Method:

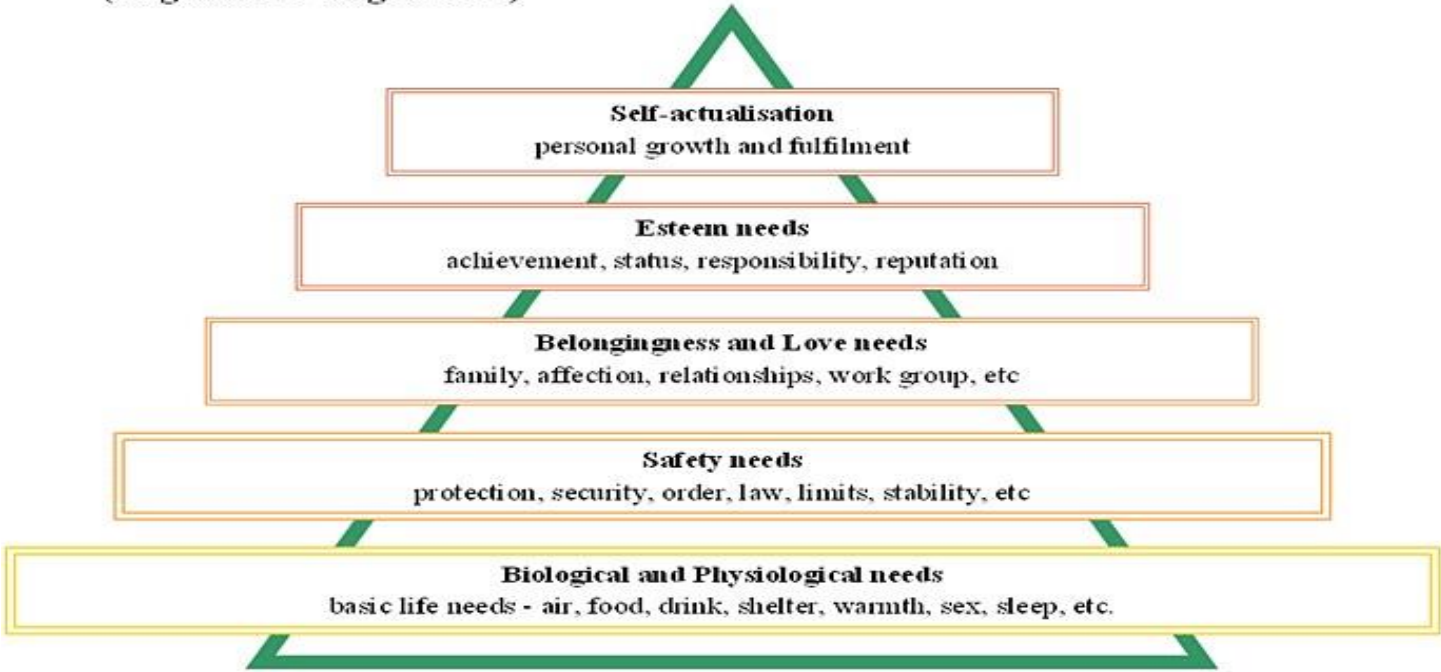
Here the teacher does not tell the students that water is necessary for plants to grow. He directly gives an experiment to the students; to put two plants in sunlight for 15 days, give water to one of them and leave the other as it is in sun. Give a result after 15 days that what happen to the plants. Now after experimenting, the students will give a conclusion that plants need water to grow, if they give such conclusion then it is definitely an inductive mode of presentation.

30. Maslow’s hierarchy of needs:

Maslow says that there are five basic needs, he made a pyramid of those needs and categorizes them as:

- 1. Biological and physiological needs.
- 2. Safety needs
- 3. Belongingness and love needs
- 4. Esteem needs
- 5. Self-actualization needs

Maslow’s Hierarchy of Needs
(original five-stage model)



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31. group learning benefits 3

Group learning will be more efficient for developing the ideas. Group learning also develops the social skills of the learners.

32. Socrates methods 2

- Socratic Method is based on dialogue generation. We can solve many problems through dialogue. Roots of problem based learning are in dialogue. Its roots are also linked with cognitive psychology. In cognitive psychology higher order skills are involved.
- The knowledge base on problem- based learning is rich and complex. Several studies done in the last few years provide strong evidence about some of the model's instructional effects. However, other studies lead to the conclusion that effects are cloudy.

33. Coach task and checker

The difference between the role of a coach and checker is that a coach tells the group what is the content and how to understand it and checker asks the group whether they have understood this or not and if not he tells the coach of the existing situation.

34. Physical Development Milestones

- **Jumping:** A teacher must give the students the opportunity for jumping exercise. They are not jumping alone; there are other students as well. So this activity is not only for physical development but social interaction as well.
- **Finger play:** When the students are learning rhymes, they must always be taught through actions. Its purpose is brain development and motor development. When the children use their motor muscles and fine muscles, as a result their motor development takes place.
- **Art activities:** All art activities are part of physical development. Art should be an essential component in pre-school education because it is directly related to physical development.

35. What is Operational learning?

Observational learning is the process of learning by watching the behaviors of others. The targeted behavior is watched, memorized, and then mimicked.

36. What is Problem-Based Learning?

The "flow" of problem-based learning:

1. Problem based learning starts from problem engagement.
2. Inquiry and investigation
3. Problem resolution
4. Problem debriefing

Basically teachers give certain problem to the students. Students engage themselves in the problems. They carry out inquiry and investigation. They define the problem. When we talk about the problem, there is some relevant and irrelevant information. If we focus on irrelevant problem we

will not be able to identify the exact problem and solve the problem. To define the problem is very important. In problem based learning students try to clarify the problem.

37. Group or Discussion advantages 5

Advantages of Group Discussion

- Learner-centred.
- It gives an opportunity to all members to give their ideas and opinion on a certain topic.
- It increases one's listening skills and confidence in speaking.
- It helps students enhance their ability to articulate and defend a position thoughtfully and respectfully.

38. Dialogue Discussion advantages 5

Dialogic Discussion

- Dialogic discussion also called as interactive discussion. This has come from Socrates method. In second script there was dialogue. First script is monologic because there was a lot of teacher control.
- In this interaction, students produce a "chain of utterances" (Nystrand, 2006) where they are able to talk without interruption or re-direction from the teacher. Students listen and respond to one another as well as the teacher. In addition, the teacher responds to the students' ideas rather than the students simply answering a series of questions asked by the teacher. Dialogic discussion promote divergent thinking. Monologic discussion promotes convergent thinking.

The long-term benefits of dialogue are improved relations and cooperation in the community, often enabling further development through social and political action. Dialogue brings slow and lasting results, change from within the community itself and brings on constructive solutions.

39. Fine motor:

It is about using hands to be able to eat, draw, dress, play, write and do many other things.

40. Intrinsic extrinsic motivation?

Expectations based upon consequences have extrinsic rewards.

- Intrinsic motivation may be defined as what motivates us to do something when we don't have to do anything. (Raffini, 1996)
- Extrinsic and intrinsic motivations are two opposing ways to motivate people. Extrinsic motivation deals with motivations that are outside of your passions, and personal selfesteem. Extrinsic motivation examples would be money, bonuses, nice cars, expensive houses, high grades in school, gold stars for athletics, etc. Extrinsic motivation is anything outside of yourself that you need to obtain or acquire to increase motivation. Intrinsic motivation is the opposite. You get paid for doing what you truly enjoy doing, nice cars and houses don't motivate you as much as your joy in work, learning, and the things that truly motivate you internally.

41. define Sequenced integration*

- Topics and units are taught independently, but they are arranged and sequenced to provide a framework for related concepts.

- Teachers arrange topics so that similar units articulate. For example, graphing units can coincide with data collection in a weather unit.

There are two subjects with colors green and yellow, they are with each other but not connected. They are fragmented but sequenced. Sequenced integration is where different subjects or disciplines are sequenced in the timetable. There is no overlapping but similar topics are taught together. So these two subjects are not linked with each other but they are being taught at the same time.

An important thing about sequenced integration was said by John Adams is:

“The textbook is not a moral contract that teachers are obliged to teach ... teachers are obliged to teach [students].”

41. define Critical thinking

- **What it isn't: Critical thinking is not necessarily being “critical” and negative. In fact, a more accurate term would be evaluating thinking.**
- **According to Robert, “critical thinking is a disciplined manner of thought that a person uses to assess the validity of something: a statement, news story, argument, research.”**
- **Beyer says that “critical thinking has two important dimensions: it is both a frame of mind and a number of specific mental operations.”** Firstly, you should have evaluative thinking, you could not evaluate a thing, unless you have evidence.

42. strategies of cooperative learning.

Cooperative learning strategies:

- **Cooperative learning strategies are:**
 1. **Think-pair-share**
 2. **Number heads together**
 3. **Jigsaw reading**
 4. **STAD**
 5. **Academic controversy**

43. What is Measurement mean

Identifying and ordering lengths and then areas, volume, weights, temperature and speeds.

44. Benefits of cooperative learning

- Higher achievement.
- Increased retention.
- More positive relationships and a wider circle of friends.
- Greater intrinsic motivation.
- Higher self-esteem.
- Greater social support.
- More on-task behavior.
- Better attitudes toward teachers.
- Better attitudes toward school.

45. what recorder do in cooperative learning

Recorder: records ideas and plans

46. Constant and variable difference

Variables are much important in scientific method. We can not apply scientific method without variables. Variables are those things that can be changed.

Constant things do not change. In the above example size of cake or quantity of sugar would be variables and oven would be a constant. Temperature would be constant.

47. Development milestone 3

Developmental milestones

Developmental milestones are set of functional skills or age-specific tasks that most children can do at a certain age range.

All children are unique they have their own developmental milestones. For example some start walking very soon some take a lot of time.

48. Integration Method 5

10 different methodologies of integration:

Fogarty and Stoehr's (1995) ten views for integrating curriculum are the most frequently used planning models in the field.

These methods are actually models of integration.

These 10 methodologies is divided into three forms.

Form- 1 – Fragmentation:

When a teacher list or rank topics, concepts and skills to systematically organize curricular priorities within each subject.

Form – 1 – Connection:

A connected methodology focuses on the details, subtitles, and interconnections within an individual discipline. Teachers help students make connections by explicitly making linkages between subject topics, skills, and concepts. Connection is a bit advance level from that of fragmentation.

Form – 1 – Nested:

Nested integration takes advantage of natural combinations. Integration is performed by overtly making connections or creating combinations. This could be accomplished in a lesson on the circulatory system by having the lesson focus on both the circulatory system

Form- 2 – Sequenced integration:

- Topics and units are taught independently, but they are arranged and sequenced to provide a framework for related concepts.
- Teachers arrange topics so that similar units articulate. For example, graphing units can coincide with data collection in a weather unit.

Form – 2 – Shared integration:

- The shared model brings two distinct disciplines together into a single focus. The shared methodology overlaps concepts as the organizer.

- The teachers of the two disciplines plan their teaching, which will take place in the individual classes together.

Form – 2 Webbed:

There are different subjects which are linked together as shown in the diagram below:

- Webbed curricula commonly use a thematic approach to integrate subject matter. Broad themes such as change, cultures, discovery, environments, interactions, and work provide a greater opportunity for teachers of various disciplines to find common topics, concepts and skills.

Form- 2 Threaded integration:

- The threaded approach to integration is a meta-curricular approach where big ideas are enlarged. This methodology threads thinking skills, social skills, study skills, graphic organizers, technology and multiple intelligences approach to

Form 2 – Integrated:

- This process blends the disciplines by finding overlapping skills, concepts and attitudes found across the disciplines. Much like the shared methodology, integration is the result of shifting related ideas out of the subject matter content.

Form 3 – Immersed integration:

It means that different disciplines are immersed under a certain umbrella.

- The immersed methodology focuses all curricular content on interest and expertise. With this methodology, integration takes place within the learners, with little or no outside intervention. Immersed learners continually make connections between their chosen topic of interest and subjects.

Form 3 – Networked integration:

- The network methodology is totally student-centered. It professes that only the learner can direct the integration process. The methodology professes that the learner knows their topic and can self-direct their focus on the necessary resources both within and across subject areas.

49. Types of discovery learning

Types of Discovery Learning: Category 1.

- **Experiments:**
- **Exploration:**
- **Simulation-based learning:**
- **Problem-based learning**
- **Inquiry-based learning**
- **Webquests**

Types of Discovery Learning: Category 2.

1. Case-based learning:
2. Incidental learning:
3. Learning by exploring/Conversing:
4. Learning by reflection:

5. Simulation-based learning:

50. Examples of discovery learning

- Are all oak leaves of the same shape? Make children aware of the diversity within leaf structure. Introduce, or reinforce, simple recording skills, and practice basic counting skills too.
- Treasure hunt
- Underground plant exploration
- Look at pictures and describe Harappa Civilization.

52. explain any three obstacles in problem-based learning 3 marks

Obstacle for PBL

The first and the most important thing is **training**. If we do not have proper training we will not be able to apply this method.

Another obstacle is school **time table**. Sometimes it will be difficult to complete the task in short period of time. Teacher may have the break time or extra class to do the task, if he may do that.

Finishing the syllabus is another big hurdle. Content is not important but process is the main thing.

Lack of resources is another obstacle. For this reason teacher needs to be innovative. He should not be the burden on the parents. Try to use the simple and available resources. Teacher can recycle the natural resources.

53. Immersed integration 5

Form 3 – Immersed integration:

It means that different disciplines are immersed under a certain umbrella.

- **The immersed methodology focuses all curricular content on interest and expertise. With this methodology, integration takes place within the learners, with little or no outside intervention. Immersed learners continually make connections between their chosen topic of interest and subjects.**

Do remember, a teacher is going to integrate the lessons with child's interest. It is not a subject specified integration but it is integration of teaching content with child's interest.

54. Snack and circle time

Snack & circle time activity: This activity is important because the children can interact with one another more easily while having their snacks. Their social interaction is more at this time and most importantly, they learn sharing here; sharing of ideas in adult life comes only if someone experiences these things in early childhood. Sharing at early ages starts with little things e.g. snacks, toffees, chips, biscuits, lunch etc.

If the children do not share these little things at early ages then they would not share their ideas in their future lives. So, snack time activity is very helpful in learning of vocabulary, sentence construction, and their social development.

55. Sub process Oriented roles

Under the **process oriented role** there are 4 sub points:

1. Gatekeeper: equalizes participation
2. Encourager: encourages members
3. Checker : helps members to check for understanding
4. Timekeeper: reminds members of progress and lack of

56. Interpreting the data

1. Interpreting data

Describing data and inferences based on them; constructing equations to represent data; relating data to hypotheses; making generalizations supported by experimental findings.

We must know how to communicate our data and interpret the data.
